SAHANA

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Profile

Application Engineer with 3+ years of experience building responsive, real-time web applications for industrial automation. Focused on debugging production issues across live customer robot fleets, supporting deployments, identifying root causes, and delivering reliable, maintainable solutions. Proficient in React.js, TypeScript, MUI, Zustand, WebRTC, and REST APIs. Delivered production-grade applications for 30+ AMRs across 5+ factories, including WebSocket-driven dashboards, and LiDAR visualization.

Technical Skills

- **Languages** : JavaScript (ES6+), TypeScript, HTML5, CSS3
- **Framework/Libraries** : React.js, Material UI(MUI), Bootstrap, D3.js, Plotly, Zustand
- **APIs & Protocols** : REST APIs, WebSocket, WebRTC
- **Visualization** : LiDAR Point Cloud (2D), Real-Time Dashboards
- **Tools** : VS Code, Vite, Git, GitHub, Grafana, Docker, Chrome DevTools,
- **Version Control** : Git, GitHub
- **Methodologies** : Agile Scrum, Component-Based Architecture, i18n (react-i18next)
- **Database** : SQL

Experience

Application Engineer – Ati Motors **June 2026 – Present**

- Investigated and **resolved production issues** across live AMR fleets by analyzing application behavior, logs, API responses, WebSocket events, and browser diagnostics, collaborating with operations teams to identify root causes
- Supported **production deployments** and post-deployment validation, including **build/deployment** issue investigation and release verification
- Built and maintained **Grafana dashboards** to track robot fleet KPIs such as operating hours, travel distance, errors, MTTR, break counts, and obstacle time.
- Updated PostgreSQL queries for Grafana dashboards and fixed data accuracy issues, including sessions crossing midnight and pre-deployment data affecting KPI totals.
- Used Docker for local application and database environments to query and troubleshoot data during development and debugging.
- Optimized React rendering using React.memo, Zustand selectors, lazy loading, and code splitting, reducing unnecessary re-renders in high-frequency real-time dashboards

Software Engineer – Ati Motors **May 2023 – May 2026**

- Built 15+ reusable React/TypeScript components for industrial dashboards, reducing development time by **25%** while improving application performance by **35%** via lazy loading, code splitting, and optimized rendering
- Embedded real-time **2D** visualization of **3D** LiDAR point cloud data into the TeleOps interface, improving operator situational awareness
- Developed real-time AMR monitoring cards with **alert prioritization** (error/warning sorting + 15-second rotation), deployed across 30+ AMRs in 3 facilities
- Connected the frontend with **3+ ERP/MES systems**, automating 20% of daily scheduling workflows
- Crafted a map-based **real-time interface** for 20+ Sherpa AMRs, reducing critical task response time by **50%**

Projects

ATI Fleet Manager

- Architected and deployed real-time operator dashboards for **30+ AMRs** across **10+ factories**, enabling supervisors to monitor **fleet status, alerts, and task execution** through sub-second **WebSocket** updates, backed by 6-month persistent **session management** ensuring uninterrupted monitoring continuity **across browser sessions** and reconnects
- Built **WebRTC-based** Teleoperation (TeleOps) system achieving 100 ms video latency and full remote AMR control, including real-time 2D LiDAR point cloud overlay for operator spatial awareness during remote navigation

- Designed **map-based real-time** UI for 20+ Sherpa AMRs with dynamic **task visualization**, **interactive zoom** to inspect **terminal lines and precise AMR positions**, and fleet coordination
- Designed a fleet **analytics dashboard** using **Chart.js** to visualize AMR KPIs, including trips, distance traveled, takt time, obstacle wait time, and dispatch wait time
- Built a **traffic signal plugin configuration** (traffic lights + patlite) with zone-based device assignment, fleet filtering, and full **CRUD management** for coordinating AMR movement at intersections

Material Tracking System (MTS)

- Led the complete frontend for an enterprise-grade Material Tracking System supporting **order processing, delivery scheduling, and AMR integration**
- Built 10+ domain-specific UI modules, including **staging area management, device configuration, WIP inventory tracking**, and approval workflows
- Established **Zustand-based** global **state management** across 10 domain stores for **real-time data consistency**
- Configured a **WebSocket** client for live AMR event updates and bidirectional communication with a **FastAPI** backend
- Built a material configuration CRUD screen with bulk CSV/Excel upload and validation support
- Built multilingual (English/Spanish) UI using **react-i18next** with locale-specific formatting
- Created **role-based** requester and approver screens for inventory approval workflows
- Connected the frontend to **PostgreSQL**-backed **REST APIs** and **MongoDB**-backed material event streams via Axios

Education

Yenepoya Institute of Technology, Moodbidri India **Aug 2018 – May 2022**

BE in Computer Science and Engineering

Relevant Coursework

JSpiders Training Institute, Bangalore India **Aug 2022 – Apr 2023**

Full-Stack Developer Training